Stage B1 Instructions and Marking Rubric

Railway digital asset development in Blender

Group submission | 100 marks | 5% of course grade | Due November 2, 2026

Project problem statement

Railway environments contain interconnected elements, including trains, platforms, station buildings, signals, lighting, vegetation, and surrounding development. A terrain model and railway tracks alone cannot communicate how these elements could form a complete environment or how people would experience it.

Your consulting team has been commissioned by a fictional municipal planning client to develop a proposed railway environment for the area around Oshawa GO Station, 915 Bloor Street West, Oshawa, Ontario L1J 8M6. The assignment study area is outlined in red in Figure 1.

For Stage B1, your task is to develop the railway and environmental assets in Blender. Use the aerial image in Figure 1 to understand the study area and plan your concept. The GIS terrain and railway track models will be provided in Lab 4; you do not need to create them for B1.

Your design task

Develop a coordinated collection of station buildings, platforms, trains, wagons or coaches, surrounding buildings, vegetation, furniture, lighting objects, a railway warning sign, and supporting infrastructure. You may introduce different architectural styles, landscaping, materials, and infrastructure designs. Existing buildings in the aerial image provide context; your proposed assets do not need to reproduce them exactly.

Choose either a passenger locomotive or leading vehicle with matching passenger coaches, or a freight locomotive with matching freight wagons. This choice identifies the train featured in your visualization; the study area remains the railway corridor around Oshawa GO Station.

Develop your own collection of assets, different from the lab example objects. Use models with visibly different geometry or design; renaming or moving an example object alone is insufficient. Use believable relative dimensions and plan a coherent layout using the aerial image. Exact placement and fit with the terrain and rails will be addressed in Lab 4.

How the stages connect

Stage B1 focuses on asset development and preparation in Blender. Unity will be introduced in Lab 4: you do not need to install it, open a Unity project, or test your models in Unity for B1. In Stage B2, you will assemble the environment, add train movement, and produce a 360-degree video. Report B will cover both stages.

Watch first. Before developing your assets, watch the example 360-degree railway video: https://youtu.be/-t2MWsJf_7o

Notice how the train, station, platforms, buildings, vegetation, and smaller objects work together. Use the example to understand why you are developing an asset collection and what you will communicate in Stage B2.

This is an academic design proposal, not an official redevelopment plan for Oshawa GO Station.

Study area at Oshawa GO Station

915 Bloor Street West, Oshawa, Ontario L1J 8M6



Figure 1. Assignment study area around Oshawa GO Station, outlined in red. Source: course-provided aerial image.

Use the red outline to define the focus of your proposed environment. For B1, annotate this aerial image or draw a simple labelled sketch showing the intended locations of your assets. This is a concept layout; a software-based scene, precise coordinates, and access to the terrain or track model are not required.

Optional reference tip. Explore the study area using Google Street View or Google Maps 3D, where available. Observe buildings, platforms, signs, vegetation, and street furniture, then use these objects as references for your own models. You may recreate or adapt what you observe while meeting the required asset list and contemporary (2026) style. Check imagery dates and record the reference links in your evidence document; an exact replica of the site is not required.

Required asset collection

Asset IDs and category	Required models and visible minimum features
A01 Locomotive or leading vehicle	One passenger or freight leading vehicle developed through Milestone 1. Show a cab with windows, main body, bogies/wheels, and coupling points.
A02 Matching coach or wagon	One separate reusable model matching A01: passenger coach with windows and doors, or an identifiable freight wagon with a suitable load body. Include wheels/bogies and coupling points.
A03 Station building	One station exterior with walls, roof, visible entrance, windows, and a station-name sign. Interior modelling is not required.
A04 Station platform	One raised platform section with a walking surface, visible thickness, and a contrasting edge strip. Size it plausibly for A01/A02; final track fit is checked in Lab 4.
A05-A06 Vegetation	A05: one tree with a trunk and foliage crown. A06: one separate shrub with a low foliage form. Use recognisably different shapes, not scaled copies.
A07-A08 Surrounding buildings	Two separate buildings in addition to A03. Each needs walls, roof, entrance and/or windows. Distinguish them by building shape and facade design; colour-only changes do not count.
A09-A10 Street furniture	A09: one bench with seat, backrest, and supports. A10: one waste bin with a body and visible opening or lid. Both must be separate reusable assets.
A11 Lighting object	One lamp post with base, pole, and light fixture. Only the physical model is required for B1; functional illumination follows in Stage B2.
A12 Railway infrastructure	One reusable railway-boundary fence segment with at least two posts and connecting mesh, rails, or panels. Keep it separate from the platform and sign.
A13 Railway warning sign	One freestanding sign with a support post and board reading “No trespassing - Railway property”. Text must be legible in the submitted close-up. A station-name sign does not count; a working signal is not required.

Submit 13 distinct assets, A01-A13, across the ten categories above. Repeated copies do not count as new assets, and one model cannot satisfy two IDs. Use these IDs in filenames, Blender object/collection names, screenshots, and the inventory. Additional assets are optional.

Work in your existing group. Every member must contribute identifiable technical work. Submit one package per group through Avenue; follow the submission time posted there.

Design consistency

Use compatible train and coach/wagon designs and believable relative sizes across the collection. A freight train may pass through the station corridor; choosing freight does not remove the station or platform requirements.

Contemporary professional appearance. Design all assets for a contemporary city and railway environment around 2026, using recent real-world references. Early-1900s locomotives, historical-themed scenes, and highly futuristic or science-fiction designs are unacceptable and will reduce the Milestone 1 mark. Older models still in contemporary service are acceptable with a recent supporting reference. Use realistic proportions, details, and materials; no cartoon styling, toy-like forms, or faces on vehicles. Simplified geometry is acceptable if the appearance remains realistic.

Originality across groups. Groups may discuss techniques and troubleshoot together, but each group must develop its own distinct asset collection. Do not submit identical or substantially similar models across groups. Renaming files, changing colours, or making minor edits is insufficient; models must differ meaningfully in geometry and design. Acknowledge any assistance and external sources.

Briefly explain your concept and train type beside the layout sketch; label planned locations with A01–A13.

Milestone 1 Digital asset development

40 marks | Based on Blender lab slides 20 to 36

Apply the lab workflow to generate your locomotive or leading train vehicle. For the remaining required assets, you may use AI generation, original Blender modelling, or suitable external models that you prepare in Blender. Identify the method and source for every asset. All required assets must pass the Milestone 2 checks.

1. Plan the collection. First choose either a freight train or a passenger train. A freight locomotive must have freight wagons; a passenger locomotive or leading vehicle must have passenger coaches. Do not combine a freight locomotive with passenger coaches, or a passenger train with freight wagons. Apply this choice throughout both stages. Sketch the intended asset locations on the aerial image within the red outline in Figure 1.

2. Develop a prompt and reference image. State the chosen service type in the prompt and describe the locomotive or leading vehicle's shape, components, colours, and materials. Create a reference image using the lab approach. Save the prompt and image. Generate the locomotive/leading vehicle and matching wagons/coaches separately as reusable assets. Specify a realistic, contemporary (around 2026), professional appearance in your prompts and choose recent real-world reference images; avoid cartoon or toy-like styling.

3. Generate the 3D geometry. Use Meshy or a suitable image-to-3D alternative. Inspect the result for recognisable railway features, complete geometry, and suitability for the intended scene. Revise the reference or regenerate if important features are missing.

4. Generate and inspect the textures. Texture the generated model using the reference image. Include PBR maps when produced by your tool. Check that the textures correspond to the model and do not hide major modelling defects.

5. Inspect and reduce mesh complexity. Record the locomotive's initial mesh statistics. Remesh or simplify it, record the final statistics using the same measure, and compare visual quality. Keep recognisable features while reducing unnecessary complexity. Explain the choice of final version; there is no universal polygon threshold for full credit.

6. Export and import. Export the generated model with its texture images and import it into Blender. FBX is the final required exchange format.

7. Complete the remaining asset categories. Develop or obtain all required replacement models. Inspect each model's appearance and complexity, and optimise where appropriate. For modified assets, retain evidence of the original and the changes. Record AI tools and external sources.

8. Save the development work. Save editable Blender files and organise each asset with its textures. Keep the selected source version and the final prepared version distinguishable so the development process can be checked.

Evidence to retain

Save your prompts, references, progress screenshots, mesh statistics, and source credits as you work. Submit the selected evidence listed under Group submission requirements; retain detailed process records for Report B.

The tools used in the lab are examples. You do not need to buy a subscription. An alternative tool or workflow is acceptable when it demonstrates the same development steps and produces usable model files. Screenshots alone cannot replace the required 3D files.

Milestone 2 Blender preparation and export

60 marks | Based on Blender lab slides 37 to 75

Complete the following checks entirely in Blender for every required asset, including imported models. Prepare each asset as a separate, reusable object and verify the exported FBX by reimporting it into Blender. Terrain and railway track modelling are outside the B1 scope.

1. Set up and save. Import the asset into Blender, place the 3D cursor at the world origin, remove unrelated default or imported objects, and save the working .blend file.

2. Set realistic scale. Use metre-based dimensions and preserve believable proportions. Use the lab's human reference or another documented dimension. Check that the train and matching wagons/coaches are compatible with each other and that platforms, doors, furniture, and buildings have sensible relative sizes. Final fit with the rail model will be checked in Lab 4.

3. Align the asset. For this assignment, use Blender Z as up and -X as the forward direction for trains and wagons. Place wheel contact points at Z = 0 and centre the train laterally on Y = 0. Place other assets with their base at Z = 0 and a sensible origin for later positioning. Verify alignment in front, side, and top views.

4. Apply the transforms. After sizing and alignment, apply the object transforms as demonstrated in the lab. Check location (0, 0, 0), rotation (0, 0, 0), and scale (1, 1, 1) for each prepared standalone mesh object. Preserve the intended geometry, dimensions, and alignment.

5. Check materials and textures. Use meaningful material assignments and make sure all image textures are available. Inspect the model in Material Preview. Replace or repair missing and incorrect assignments. A solid-colour material is acceptable where it suits the object.

6. Check UV mapping. For image-textured assets, inspect how faces map to the texture. You generally, do not need to do anything here; but, you need to document in your final report that you checked and understood the concept.

7. Check face orientation. Use Face Orientation to inspect visible exterior surfaces. Recalculate outward-facing normals and repair major reversed faces that affect appearance. Document any small unresolved areas and their effect.

8. Name and organise the components. Give descriptive, consistent names to objects, meshes, materials, textures, and files. For example: StationBuilding, StationBuilding_Mesh, and StationBuilding_Mat.

9. Export the prepared asset. **Export each asset separately as FBX using the six settings in the UPDATED Blender lab slide 72: (1) Path Mode: Copy; (2) Embed Textures: ON; (3) Selected Objects: ON; (4) Object Type: Mesh; (5) Apply Unit: ON; (6) Apply Transform: OFF.** Check these settings before every export. The Apply **Transform export checkbox must remain OFF**; this is separate from applying object transforms in Step 4. Include the referenced texture images in your submission, even when embedded in the FBX. Each FBX must contain exactly one asset ID and only its own component parts. Do not attach, join, or export unrelated objects with it. For example, a locomotive may include its wheels and couplers, but no rails, ground, platform, wagons, or other scene objects. Export every coach/wagon, building, platform, and other required asset separately, never as a combined scene.

10. Verify the exported file. Import the FBX into a fresh Blender file and check geometry, dimensions, orientation, material appearance, and texture availability. Resolve significant problems before submission. Record the result for every required asset; use the final exported files in Stage B2.

Milestone 3 and its advanced techniques remain optional.

Group submission requirements

Submit one ZIP per group through Avenue by the posted time. Name it GroupXX_StageB1.zip, replacing XX with your group number. Include the actual model and texture files, not only screenshots, download links, or cloud references.

Folder or file inside the ZIP	Required contents
01_Blender	One final editable .blend file per ID, e.g. A13_WarningSign.blend. Include all parts of that asset and pack or supply its images. Store development versions in 03_DevelopmentEvidence. **13 final Blender (.blend) files**, one for each asset ID **A01–A13**.
02_Exports	One separate FBX per asset ID, containing only that asset and its own parts. Use a subfolder per ID, e.g. A13_WarningSign/A13_WarningSign.fbx and A13_WarningSign/Textures/. Include referenced images even when embedded. No combined scenes or unrelated attached objects. **13 FBX (.fbx) files,** one for each asset ID **A01–A13**.
03_DevelopmentEvidence	A01 prompt, reference image, generated/textured previews, and before/after optimisation evidence. Include original-versus-modified evidence where applicable. Label files by asset ID.
GroupXX_StageB1_Evidence.docx	Brief illustrated evidence covering the items below. No separate preparation checklist or full report is required for B1.
README.txt	Group/company name, members, Blender version, chosen passenger/freight type, and any missing files or known problems.

Contents of the evidence document

1. Group and concept. List your group and members, selected train type, and a labelled concept sketch showing A01–A13.

2. Asset inventory. List A01–A13 with asset name, responsible member, method/source, dimensions in metres, and Blender/FBX paths. Include a labelled final preview of each asset; contact sheets are acceptable. Show A13's sign text clearly.

3. Milestone 1. Show the A01 prompt, reference image, generated/textured previews, and before-and-after mesh statistics using the same measure. Briefly explain the selected result and changes to external models.

4. Milestone 2. Include selected A01 screenshots showing scale/alignment and applied transforms, plus the export settings from updated slide 72. Confirm that all 13 FBX files were reimported and checked; briefly list unresolved issues. No separate checklist is required.

Before uploading

Extract the ZIP into a new folder and open the submitted files there. Confirm all 13 IDs have a separate Blender file, FBX export, required textures, inventory entry, and preview. Paths must work without files elsewhere on your computer.

Keep the B1 evidence brief. Retain detailed screenshots, working notes, and explanations of your modelling and preparation process for Report B in Stage B2. A full report is not required for B1.

Stage B1 marking rubric

Maximum 100 marks | Weight 5% of the final course grade

Milestone	Marks	Full credit standard
Milestone 1 Digital asset development	40	Provides A01–A13 with the listed visible features and realistic, contemporary (circa 2026) city-planning appearance. Train and coach/wagon types match. Demonstrates A01 prompting, reference-image creation, geometry, texturing, mesh reduction/comparison, export, and Blender import. Methods, sources, contributions, and required files/evidence are traceable by asset ID.
Milestone 2 Blender preparation and export	60	Prepares all assets with appropriate scale, alignment, transforms/naming, materials/UVs, normals, slide 72 export settings, and fresh-file reimport. Supplies 13 Blender files and 13 separate FBX exports with required textures, labelled previews, selected A01 screenshots, and confirmation of reimport checks. Evidence and files agree; the package is ready for Stage B2.

Fixed score performance scale

Select one fixed performance level for each milestone: 100%, 90%, 75%, 60%, 35%, or 0% of its allocated marks. Do not assign scores between these levels. Judge the severity and practical effect of deficiencies across the required collection.

Performance level	Score	Decision rule
Meets Requirements	100%	The full-credit standard is met. Any imperfection is negligible and needs no meaningful correction.
Needs Limited Improvement	90%	The work is essentially complete and technically sound. A small, identifiable correction or addition is needed.
Needs Moderate Improvement	75%	The core requirement is met, but several issues or omissions need revision. The work remains usable.
Needs Major Improvement	60%	Important elements are present, but significant errors or omissions materially weaken the milestone.
Needs Substantial Improvement	35%	A meaningful attempt is evident, but major deficiencies leave much of the milestone incomplete or unusable.
Not Completed or No Meaningful Evidence	0%	The milestone is missing, not attempted, or lacks enough meaningful evidence to assess.

Use the listed asset features and milestone steps to judge performance. Keep the 40/60 milestone weighting; avoid duplicate deductions for the same deficiency. Cite the asset ID, evidence, impact, and correction in feedback. All 13 assets are required; Milestone 3 remains optional.

Quick marking sheet

Group _____ Assessor _____ Date _____ Give a short reason for each milestone below full marks.

Criterion	Max	100%	90%	75%	60%	35%	0%	Awarded	Reason or specific deficiency
Milestone 1	40	40	36	30	24	14	0	_____	_____
Milestone 2	60	60	54	45	36	21	0	_____	_____
Stage B1 total	100							_____	

Course-grade contribution = (Stage B1 mark / 100) x 5 percentage points.